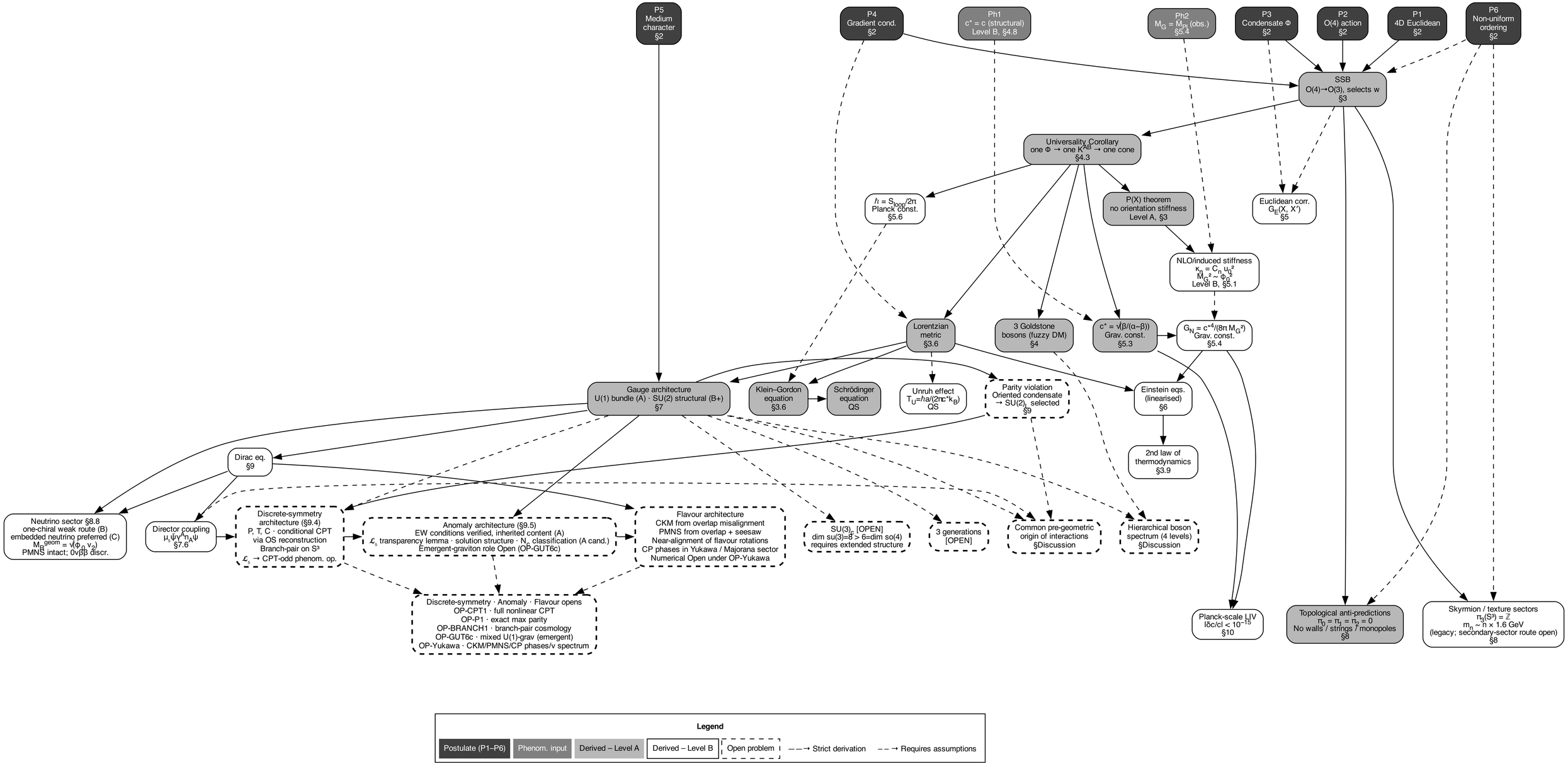
P5
Medium character §2
P4
Gradient cond. §2
Ph1
c* = c (structural) Level B, §4.8
Ph2
M_G = M_Pl (obs.) §5.4
P3
Condensate Φ §2
P2
O(4) action §2
P1
4D Euclidean §2
P6
Non-uniform ordering §2
SSB
O(4)→O(3), selects w §3
Universality Corollary
one Φ → one K^AB → one cone §4.3
ħ = S_loop/2π
Planck const. §5.6
P(X) theorem
no orientation stiffness
Level A, §3
Euclidean corr.
G_E(X, X′) §5
NLO/induced stiffness
κ_n = C_n u_*^2
M_G^2 ~ Φ_0^2
Level B, §5.1
Lorentzian metric §3.6
3 Goldstone bosons (fuzzy DM) §4
c* = √(β/(α−β))
Grav. const. §5.3
G_N = c*^4/(8π M_G^2)
Grav. const. §5.4
Gauge architecture
U(1) bundle (A) · SU(2) structural (B+) §7
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Unruh effect
T_U=ħa/(2πc*k_B) QS
Parity violation
Oriented condensate
→ SU(2)_L selected §9
Einstein eqs. (linearised) §6
2nd law of thermodynamics §3.9
Dirac eq. §9
Neutrino sector §8.8
one-chiral weak route (B)
embedded neutrino preferred (C)
M_D^geom = y(Φ_0, v_2)
PMNS intact; 0νββ discr.
Director coupling
μ_n ψγ^A n_A ψ §7.6
Discrete-symmetry architecture (§9.4)
P, T, C · conditional CPT
via OS reconstruction
Branch-pair on S³
ℒ_5 → CPT-odd phenom. op.
Anomaly architecture (§9.5)
EW conditions verified, inherited content (A)
ℒ_5 transparency lemma · solution structure · N_c classification (A cand.)
Emergent-graviton role Open (OP-GUT6c)
Flavour architecture
CKM from overlap misalignment
PMNS from overlap + seesaw
Near-alignment of flavour rotations
CP phases in Yukawa / Majorana sector
Numerical Open under OP-Yukawa
SU(3)_c [OPEN]
dim su(3)=8 > 6=dim so(4)
requires extended structure
3 generations [OPEN]
Common pre-geometric origin of interactions §Discussion
Hierarchical boson spectrum (4 levels) §Discussion
Discrete-symmetry · Anomaly · Flavour opens
OP-CPT1 · full nonlinear CPT
OP-P1 · exact max parity
OP-BRANCH1 · branch-pair cosmology
OP-GUT6c · mixed U(1)-grav (emergent)
OP-Yukawa · CKM/PMNS/CP phases/ν spectrum
Planck-scale LIV
l0c/cl < 10^−15 §10
Topological anti-predictions
π_0 = π_1 = π_2 = 0
No walls / strings / monopoles §8
Skyrmion / texture sectors
π_3(S³) = ℤ
m_n ~ n × 1.6 GeV
(legacy; secondary-sector route open) §8
Legend
Postulate (P1–P6)
Phenom. input
Derived – Level A
Derived – Level B
Open problem
Strict derivation
Requires assumptions